EXAMPLE 6.7

Light travelling through transparent oil enters in to glass of refractive index 1.5. If the refractive index of glass with respect to the oil is 1.25, what is the refractive index of the oil?

Solution

Given, n_{go} =1.25 and n_g =1.5

Refractive index of glass with respect to oil,

$$n_{go} = \frac{n_g}{n_o}$$

Rewriting for refractive index of oil,

$$n_o = \frac{n_g}{n_{go}} = \frac{1.5}{1.25} = 1.2$$

The refractive index of oil is, n_o = 1.2

6.4.5 Apparent depth

It is a common observation that the bottom of a tank filled with water appears to be raised when seen from air medium above as shown in Figure 6.19(a). An equation could be derived for the apparent depth when viewed in the near normal direction.

Light from the object *O* at the bottom of the tank passes from denser medium (water) to rarer medium (air) to reach our eyes for viewing the object as shown in Figure 6.20 (b). It deviates away from the normal in the rarer medium at the point of incidence *B* as shown in Figure 6.20 (c). The refractive index of the denser medium is n_1 and that of rarer medium is n_2. Here, $n_1 > n_2$. The angle of incidence in the denser medium is *i* and the angle of refraction in the rarer medium is *r*. The lines *NN′* and *OD* are parallel. Thus, the angle ∠*DIB* is also *r*. The angles *i* and *r* are very small as the diverging light from *O* entering the eye is very narrow. The Snell's law in product form for this refraction from equation (6.19) is,

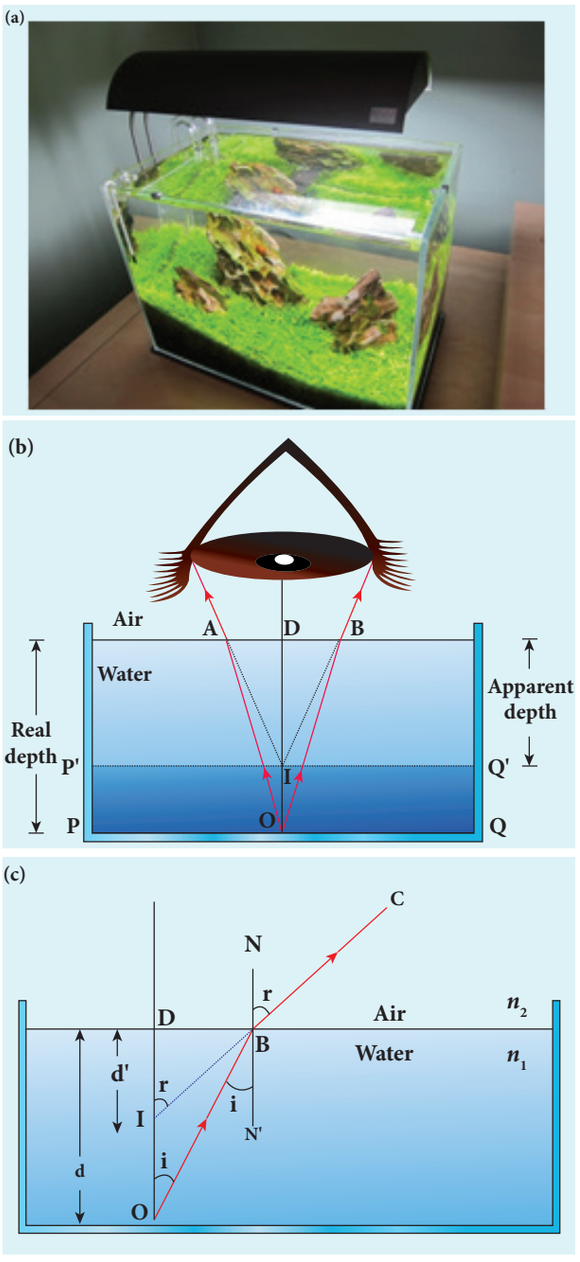


Figure 6.20 Apparent depth

$$n_1 \sin i = n_2 \sin r$$

As the angles *i* and *r* are small, we can approximate, $\sin i \approx \tan i$ and $\sin r \approx \tan r$.

$$n_1 \tan i = n_2 \tan r$$

In triangles Δ*DOB* and Δ*DIB*,

$$\tan i = \frac{DB}{DO} \text{ and } \tan r = \frac{DB}{DI}$$

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